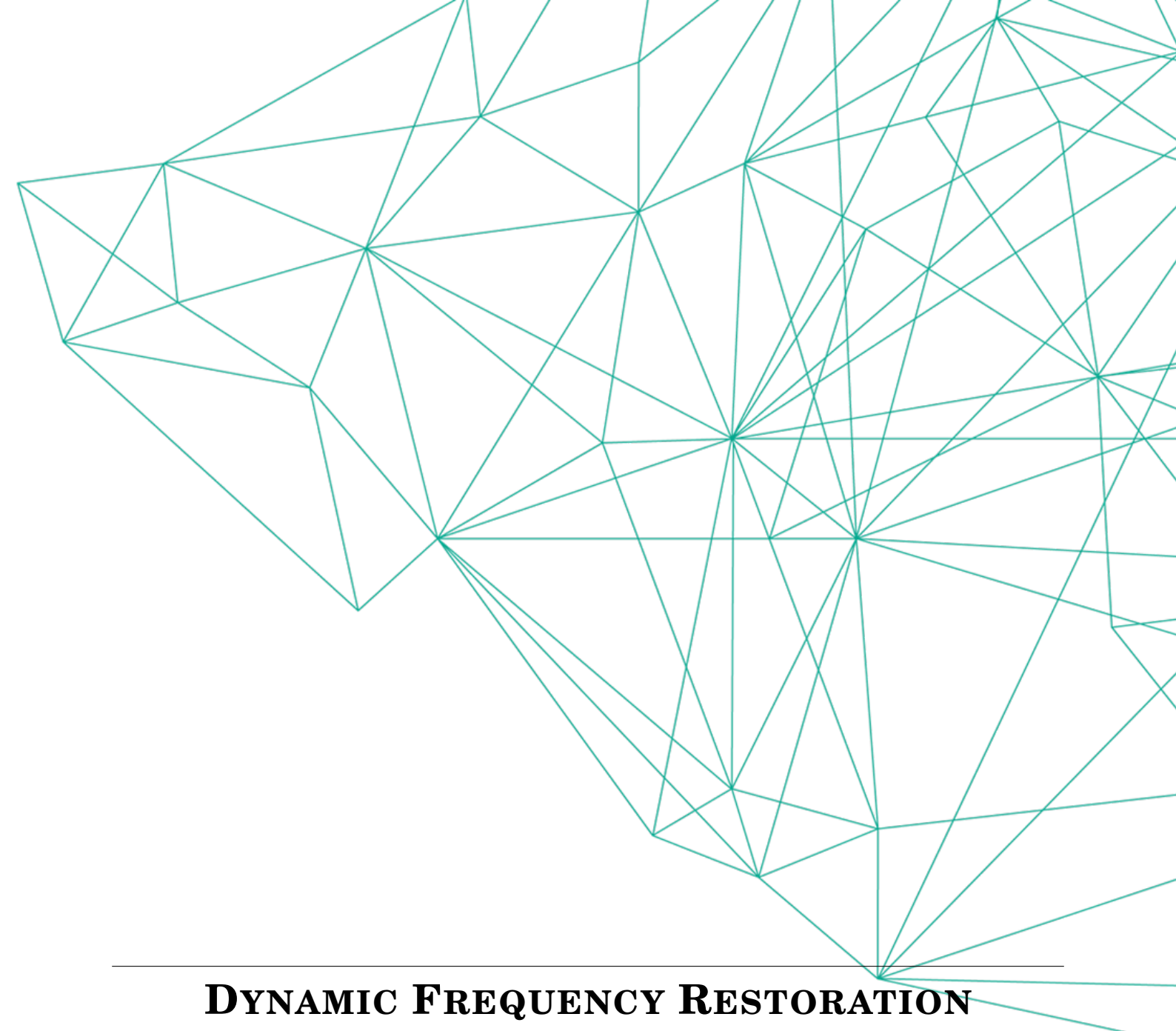




DYNAMIC FREQUENCY RESTORATION RESERVE DIMENSIONING IN THE NORDICS: STATUS QUO AND BEYOND

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1 Introduction

The increasing penetration of renewable energy sources (RES), such as wind and solar, introduces significant challenges for maintaining power system balance. Unlike conventional synchronous generation, RES are inherently variable and uncertain, leading to forecast errors and intermittent production. These characteristics amplify system imbalances and create a need for more flexible and cost-efficient reserve dimensioning strategies.

Traditionally, Transmission System Operators (TSOs) have relied on *static dimensioning* to determine reserve requirements, sizing reserves to cover the largest single contingency within a bidding zone (the N-1 security criterion) and applying these fixed values over long time horizons. The conceptual foundations were laid by early reserve-margin planning studies for mixed wind–hydro–thermal systems [1] and refined through methodologies for isolated power systems [2], with reliability-oriented assessment frameworks such as system well-being analysis providing a complementary perspective on adequacy [3]. While these approaches work well in systems dominated by predictable generation, static dimensioning fails to capture the time-varying nature of imbalances caused by high RES penetration, network contingencies, and congestion.

To address these limitations, research has progressively turned to probabilistic methods for reserve sizing. Early work quantified how growing wind penetration inflates reserve demands [4] and proposed estimation procedures for spinning reserve under significant wind generation [5]. A natural next step was to drive the reserve level directly from probabilistic wind-power forecasts, linking forecast uncertainty to operating reserve requirements [6], and to embed this uncertainty inside a joint reserve-scheduling and N-1 security framework [7]. The breadth of these methods is surveyed in [8]. More recent contributions extend the probabilistic view to zonal reserve requirements with explicit deliverability [9] and to scalable sizing procedures designed for grids with very high renewable penetration [10]. These approaches improve reliability but remain insufficient when applied within deterministic procurement frameworks. In parallel, a complementary line of work has examined the market and network implications of reserve allocation. Co-optimisation of energy and reserves has been formulated with contingency-based zonal pricing [11], and extended to incorporate post-deployment transmission constraints in zonal ancillary-service markets [12]. Reserve requirements have also been shown to be an effective lever for managing intra-zonal congestion [13], underscoring that reserve sizing is tightly coupled to both pricing and grid operation.

Against this backdrop, *dynamic dimensioning* has emerged as a promising methodology that adjusts reserve procurement on an hourly basis, considering real-time system conditions and risk preferences rather than relying on static rules. The economic case for cross-border exchange and sharing of generation reserves between TSOs has been analysed in [14], and a concrete operational realisation of these ideas is presented in the recent N-Side report on dynamic dimensioning with the Nordic TSOs [15]. The purpose of this paper is to present this concept, discuss the challenges and opportunities associated with its implementation in a Nordic setting, and explore potential methodological improvements and long-term perspectives for coordinated reserve sizing.

The remainder of this paper is structured as follows: Section 2 introduces the foundational concepts of ancillary services and balancing markets in a European and Danish context. Section 3 motivates in detail the potential limitations of a static reserve procurement, and how a dynamic (and multi-area) strategy can address these. Section 4 describes Energinet’s current work with dynamic dimensioning and highlights key open questions. This is followed by Section 5 which discusses potential methodology extensions.

2 Technical Preliminaries of Dynamic Dimensioning

To provide context for dynamic dimensioning, this section introduces the necessary concepts of the European power market structure. Balancing aspects are the primary focus, and therefore *Ancillary Services* will be covered thoroughly.

2.1 The European Power System and Ancillary Services

The European power grid operates under a zonal structure, where each Transmission System Operator (TSO) is responsible for maintaining system stability within each zone. These zones, commonly referred to as bidding zones and sometimes as Load Frequency Control (LFC) areas, may span entire countries, such as the French and German bidding zones, or be subdivided into smaller regions, as in Denmark (DK1 and DK2) and most Nordic countries. Another common term is *Control Area* which is the geographical area that each TSO controls. In most cases, there is a single TSO per country (and therefore all bidding zones within a country will fall under one control area), except for the case of Germany which has a single bidding zone but four TSOs, and thus four control areas.

The European system is also separated into two synchronous areas: Continental Europe and the Nordic synchronous area. Denmark is unique in belonging to both. DK1 (Jutland and Funen) is part of Central Europe, while DK2 (Zealand) is part of the Nordic synchronous area.

To mitigate deviations from scheduled power generation and consumption, arising from the Day-Ahead wholesale market, European TSOs procure ancillary services to maintain real-time balance. The ancillary services discussed in this paper are limited to *balancing reserves* as these are targeted with dynamic dimensioning. Other services, such as voltage support and black-start capability, fall outside the scope.

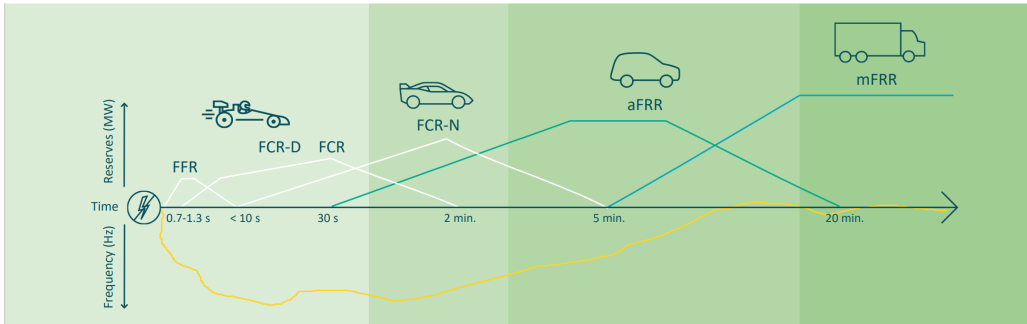


Figure 1: Overview of the different Ancillary Service products used for Grid Balancing. The figure is adapted from Energinet’s *Outlook for Ancillary Services* report [16].

2.1.1 Balancing Reserves

Balancing reserves in Europe are grouped into three categories: primary, secondary, and tertiary. In Continental Europe, these correspond to Frequency Containment Reserve (FCR) (primary), Automatic Frequency Restoration Reserves (aFRR) (secondary), and Manual Frequency Restoration Reserves (mFRR) (tertiary). The Nordic system uses the same structure but includes additional products for primary reserves: Frequency Containment Reserve - Normal (FCR-N), Frequency Containment Reserve - Disturbance (FCR-D) to handle ordinary operating conditions, and Fast Frequency Reserve (FFR) to address low-inertia conditions. The different ancillary service products and the order of activation in case of a contingency is illustrated in Figure 1.

Primary reserves respond directly to frequency deviations and act as contingency measures. They provide limited energy but stabilize large frequency drops. Secondary and tertiary reserves, collectively referred to as Frequency Restoration Reserves (FRR), are more energy-intensive and handle both smaller imbalances and major contingencies. They are used for both positive imbalance (downregulation) and negative imbalance (upregulation). aFRR and mFRR have full activation times of 5 and 15 minutes, respectively. This paper focuses on these two products, as these are the products that are relevant to dynamic dimensioning.

For a comprehensive overview of ancillary services and future trends, see Energinet's biannual report *Outlook for Ancillary Services* [16]. Technical requirements for each product are detailed in Energinet's *Tender Conditions* [17].

Entities providing ancillary services must be certified as Balancing Service Provider (BSP)s and ensure that each unit is prequalified for the relevant product. Prequalification verifies compliance with technical requirements for activation and delivery. Details on Energinet's prequalification process are available in [18].

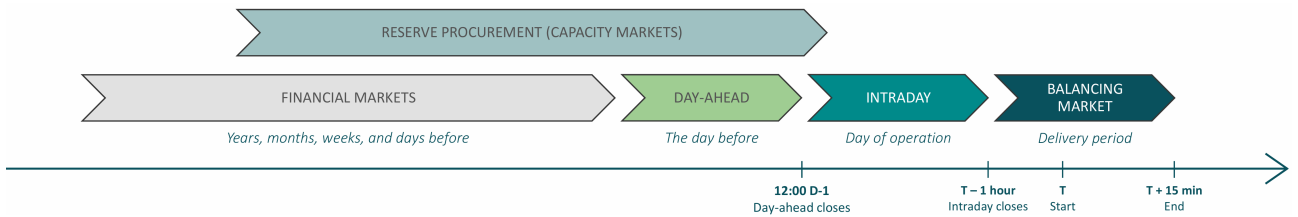


Figure 2: Overview of the market timelines in the European system. Financial markets, such as Power Purchase Agreements, are settled long-term. Reserve procurement through the capacity market is handled mostly on the day before operation, before the day-ahead wholesale market. The intraday market is then open until 1 hour before operation, and the energy activation markets for aFRR and mFRR run from one hour before operation until 15 minutes before operation.

2.2 Balancing markets

Unlike the Day-Ahead Market, managed by dedicated power exchanges such as Nord Pool and EPEX, ancillary services are procured directly by TSOs through the Capacity Market (CM) and Energy Activation Market (EAM), commonly referred to as the *balancing markets*. Capacity markets secure reserves in

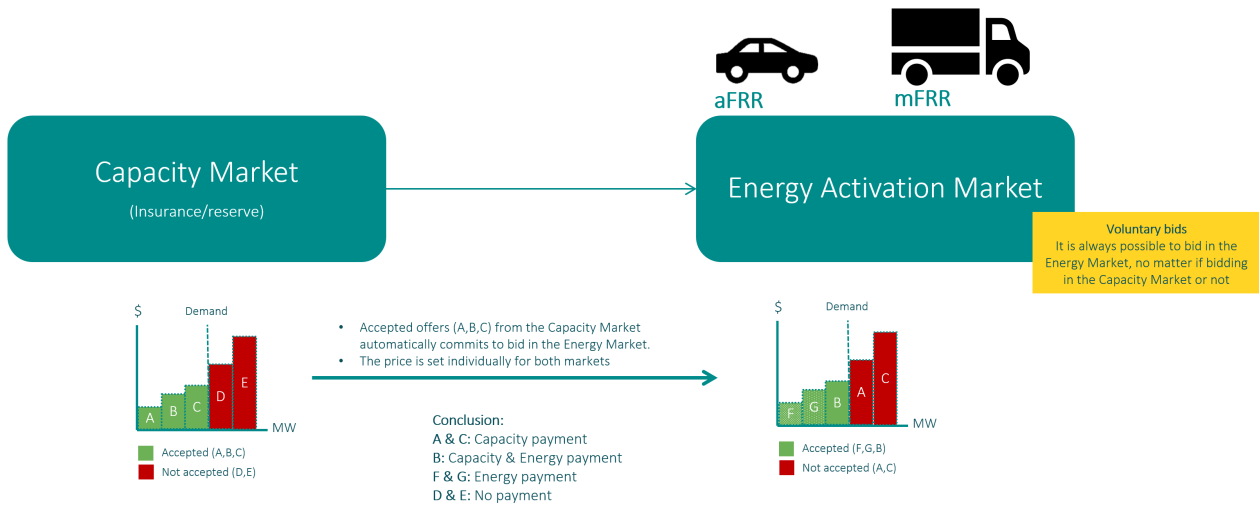


Figure 3: The bids from the capacity markets are obligated to participate in the Energy Activation Markets, but voluntary bids can also bid into the EAM. The example in the figure shows a case where the bid that was contracted through the capacity market does not get accepted in the energy activation market.

advance, while energy activation markets determine real-time deployment based on system imbalances. A timeline showing the different markets and their gate closure time can be seen in Figure 2.

2.2.1 Capacity Markets

The capacity market is held once a day for the coming 24-hour period in two separate markets for mFRR and aFRR. The BSPs submit bids for single hours or consecutive blocks, specifying *volume* and *price*. The capacity market clearing follows a *pay-as-cleared* principle, where the marginal accepted bid sets the price for *all* accepted bids. The accepted bids receive a *capacity payment* for being available, regardless of whether they are activated in real time or not. Energinet covers these costs, which are socialized through system tariffs paid by end consumers. Because demand in capacity markets is inelastic, even small changes in procurement volumes can strongly influence clearing prices.

2.2.2 Energy Activation Markets

Accepted bids from the capacity market become *contracted bids*, which must be available during the contracted hour and participate in the EAMs for the entire hour (multiple EAMs are held within each hour). The contracted bids can choose another price when entering the EAM and must participate with the same or, if possible, higher volume than they bid in the capacity market.

Activation markets also use pay-as-cleared pricing, but costs are allocated through an *imbalance price* mechanism based on the *polluter-pays* principle. Participants causing imbalances bear the associated costs, while those supporting system balance are rewarded [19]. The imbalance price is determined as the weighted volume of aFRR and mFRR activation prices.

Common European EAM platforms for aFRR and mFRR have been developed, called Platform for the International Coordination of Automated Frequency Restoration and Stable System Operation (PICASSO)

and Manually Activated Reserves Initiative (MARI), respectively. Energinet and Fingrid joined PICASSO in 2024, while the other Nordic countries will be joining in the coming years. The Nordic TSOs have developed their own common platform for automatic activation of mFRR, as part of the Nordic Balancing Model, and are planning to connect that to the MARI platform by the start of 2027 [20].

2.2.3 Voluntary bids

The EAM is open for BSPs to bid in with all the pre-qualified capacity they have, regardless of whether they were contracted through the capacity market or not. Such non-contracted bids are called *voluntary bids* and are not obligated to participate. The relationship between the capacity market, the EAM and voluntary bids is illustrated in Figure 3.

Voluntary bids matter for dimensioning because they allow Energinet to reduce contracted aFRR/mFRR volumes in periods where many Voluntary Bids (VB)s are expected, leveraging the different financing of the two markets to lower procurement cost. The trade-off is security of supply: if forecasted voluntary capacity does not materialise, the EAM may be short, since no commitment was secured up front.

2.3 Principles of Reserve Dimensioning

Reserves are dimensioned to cover two sources of imbalance, *Reference Incidents* and *Normal Imbalances*, while exploiting two opportunities to reduce the required volume, *Netting* and *Sharing*. These four concepts are introduced below.

2.3.1 Reference Incidents

A Reference Incident (RI) is the largest possible outage within a single bidding zone, e.g. the largest producing unit and the largest consuming unit. This is typically the largest generator or interconnector to another bidding zone. The RI is only known *after* the day-ahead market clears, so until that point, Energinet must rely on a forecast of the RI. In practice it is most often the interconnector to Great Britain, called Viking Link, that sets the RI level for DK1, while it is the interconnector between DK1 and DK2 (Great Belt) that sets the RI level for DK2.

2.3.2 Normal Imbalances

The Normal Imbalances (NI) is the imbalance that occurs in normal operation of the grid, due to a mismatch from the dispatch schedule that was created after clearing of the Day-ahead market (with potential adjustments to the schedule from trading in the intraday market). In traditional power systems dominated by firm capacity, most of the deviations stemmed from deviations in the consumption, since household consumption etc. are harder to control firmly. In systems dominated by intermittent renewables, like Denmark with high wind and solar penetration, the production side is now also contributing heavily to the normal imbalance due to forecast errors of available production, as well as deviations within Market Time Unit (MTU)s.

2.3.3 Sharing of Reserves

The European TSOs are allowed to share their reserves to cover a potential RI in a neighbouring bidding zone. This concept is only allowed for major outages and cannot be used to cover NI in other zones. An example of a sharing agreement between two zones can be seen in Figure 4. This can be used to reduce the reserve need of a given zone, given the assumption that multiple RIs are highly unlikely to happen at the same time, thus the reserves to cover the RI are not needed at the same time for both zones.

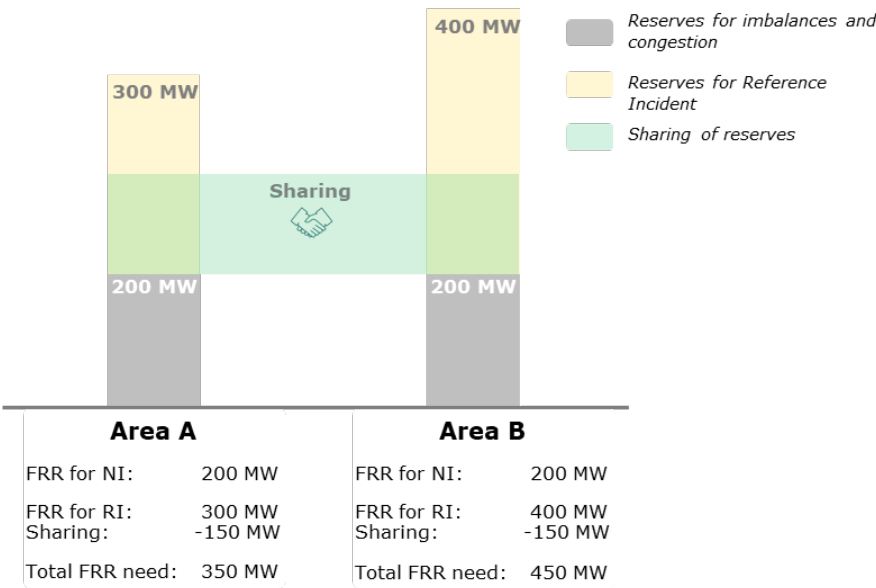


Figure 4: A simple example of how sharing affects reserve dimensioning. With a Reference Incident of 300MW in area A and 400MW in area B, they assume that a reference incident will not happen at the same time. With an expected ATC between the two zones of 150MW (in both directions) the two zones can now lower their reserve procurement by this amount. In practice, if a reference incident were to occur in area A, they will rely on the reserves bought in area B to support up to 150MW. Note that if a reference incident does occur, and reserves are activated, the associated costs will be covered by zone A.

2.3.4 Netting of imbalances

While the reserves dedicated to cover RIs are not allowed to be used to help another zone to cover their NI, TSOs can utilize *netting* of opposite imbalances between two zones to balance their grid, instead of activating ancillary services. The concept of netting is shown in Figure 5. Simply put, if the imbalances between two bidding zones are opposite, the two zones can level out their imbalances if they are connected. This is only possible if there is enough capacity available on the interconnector between the zones.



Figure 5: Example of netting between two bidding zones. The imbalance of area A is positive, while being negative in area B. The ATC between the two zones is 50MW (in the direction from A to B). This allows the imbalances to be 'netted' to +50MW in area A, and -50MW in area B. The alternative would be to make local activations in both area A (down-regulation) and area B (up-regulation).

2.3.5 Available Transfer Capacity

For sharing and netting to be an available option for TSOs, they must ensure that there is enough *Available Transfer Capacity* on the interconnector connecting two bidding zones. The interconnectors have a fixed capacity, that can be utilized in the different markets. The TSOs can book up to 10% of the transfer capacity of an interconnector to make room for cross zonal activations in the capacity market, i.e. buying reserves in another zone. The remaining capacity is then available to the day-ahead market to allow cross-zonal power flow. The *residual* capacity on the interconnectors, also considering further trades in the intraday market, is the Available Transfer Capacity (ATC) that can be used in the balancing time frame, i.e. utilized for sharing and netting purposes.

2.4 FRR sizing

The FRR volume to procure is built up bottom-up, as illustrated in Figure 6: the reference incident forms the bulk of the requirement (reduced by sharing opportunities), the expected normal imbalance is added on top (reduced by netting), and finally the expected voluntary bids in the EAM are subtracted to give the final FRR need. Special regulation, which covers e.g. local congestion management, is out of scope of this work and will not be discussed further. Each component can be either forecasted on an hourly basis to determine the FRR need *dynamically*, or fixed at a value calibrated to a chosen historical percentile in a *static* setting.

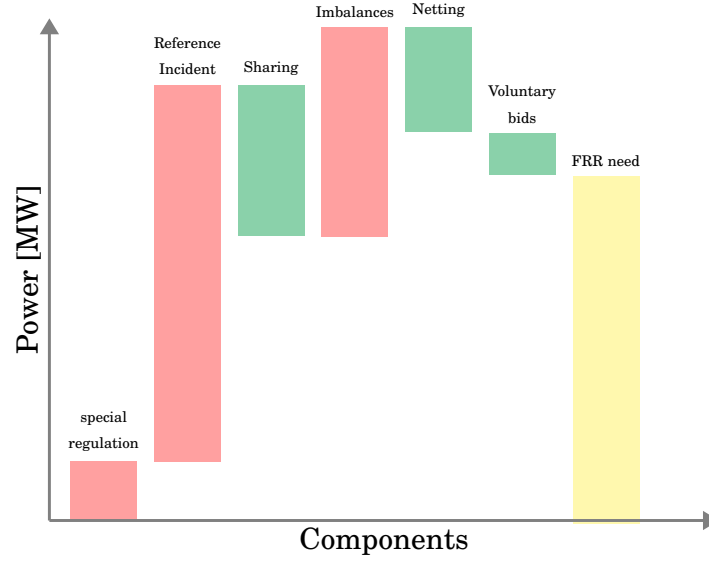


Figure 6: The components that are used to determine the reserve needs. Red boxes indicates additions to the reserve needs, while green boxes are sources for reductions of FRR need. The final column is the remaining FRR need, which is used as a target value to train a probabilistic forecasting model.

3 Benefits of a dynamic dimensioning approach

This section builds on the concepts introduced previously to motivate the topic of dynamic dimensioning. It explains why dynamic dimensioning is important, what approaches are currently applied, and what Energinet aims to achieve with its methodology. The structure is as follows: first, the historical methodology of reserve sizing is outlined; next, emerging approaches in Europe are discussed; finally, the section highlights why the Nordic context represents a particularly interesting case for investigation.

3.1 Why Dynamic Dimensioning?

As noted in Section 2, the prevailing practice has been to set reserve volumes *statically*: aFRR and mFRR are calculated annually to cover either the RI of the bidding zone or the 99th percentile of historical NI, whichever is greater. This methodology is described in the European System Operation Guideline (SOGL) [21], which provides the legal framework for the European TSOs.

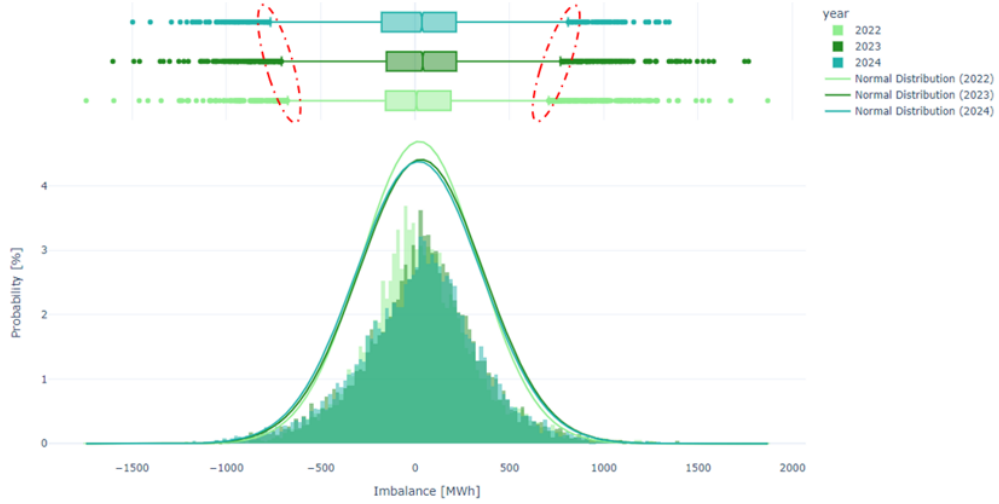


Figure 7: The change in normal imbalance distributions from 2022 to 2024. Periods with a larger normal imbalance can be seen to occur more frequently, as the distributions get wider, detailed in the boxplot. Datasource: Energinet

However, recent years have seen increasing volatility in NI in Denmark [16], as illustrated in Figure 7. This trend is primarily driven by forecast errors associated with intermittent renewable generation such as wind and solar. Greater volatility amplifies the importance of considering imbalance uncertainty when dimensioning reserves and underlines the growing need for ancillary services to manage normal imbalances effectively.

A static approach is therefore no longer adequate. The goal of dynamic dimensioning is to forecast the components of Figure 6 on an hourly basis and combine them into a reliable reserve level for procurement in the capacity market, ensuring security of supply while minimizing redundant reserves [22]. Beyond reliability, the dynamic method also has potential to be more cost-effective: Figure 8 illustrates how even a modest reduction in the procured volume can decrease procurement cost significantly, owing to the inelasticity of capacity-market demand.

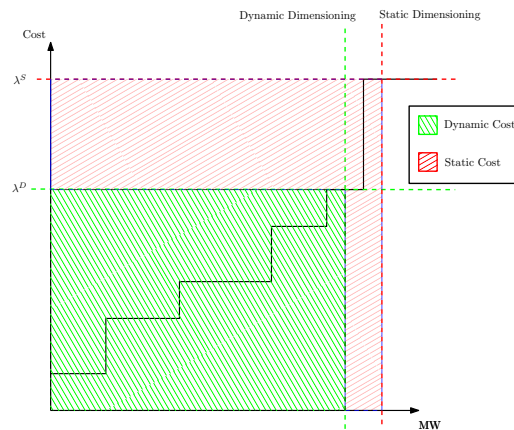


Figure 8: Illustrative example of how reducing the reserve needs for a given hour can reduce the cost of buying reserves. This could be the case if a dynamic model would estimate a lower need than a static method for a given hour.

Other TSOs in Europe have also adopted dynamic FRR dimensioning principles. For example, ELIA

developed a proof-of-concept using Monte Carlo simulation and machine learning to optimize reserve procurement based on short-term system conditions [23]. This approach is now operational in Belgium. Similarly, the Dutch TSO TenneT in the Netherlands has implemented a methodology combining stochastic imbalance estimation with deterministic analysis of the largest contingency in a probabilistic framework [24]. Energinet introduced its own dynamic dimensioning methodology for FRR in January 2024 [25], which places particular emphasis on Nordic netting and reserve sharing capabilities as well as voluntary bids in the EAM within a probabilistic framework.

3.2 Benefits of Multi-Area reserve dimensioning

The SOGL specifies reserve dimensioning rules at the LFC-block level, meaning imbalances and reference incidents should be considered for the entire Nordic synchronous area, as this is one LFC-block. In practice, applying these guidelines strictly would result in very low reserve levels. The Nordic TSOs agree that the SOGL target is insufficient, and they instead buy reserves to cover the reference incident for each of their LFC-areas. However, with proper coordination, sharing and netting can reduce overall FRR reserve capacity needs while maintaining high security of supply. Treating dynamic dimensioning as a *multi-area* problem enables TSOs to leverage sharing of surplus reserves that are unlikely to be needed simultaneously, assuming only one RI occurs at a time. Recognizing this opportunity, Nordic TSOs have amended the SOGL [26] to commit to a common Nordic methodology that supports dynamic, multi-area reserve dimensioning.

3.3 Fairness of reserve sharing

Within a multi-area framework, fairness in reserve sharing is critical. Without transparent allocation rules, benefits may be unevenly distributed among TSOs. This challenge can be modeled as a resource allocation problem, where each TSO seeks to minimize its own reserve procurement while meeting NI and RI requirements. Game theory can in this context be used to investigate the fairness of such problems [27].

From a welfare-economic perspective, reserves should ideally be placed where they provide the greatest system-wide benefit, i.e. it will maximize the sum of utility in the system. However, this may disadvantage well-connected TSOs, which could bear a disproportionate share of procurement costs. This creates an *efficiency-fairness trade-off* [28]. Potential ways of tackling this issue include: (1) *fairness-by-design*, where allocation rules enforce fairness in each daily dimensioning process (each game); (2) *dynamic fairness*, ensuring convergence to fair outcomes over time (across many games); and (3) complete welfare maximization but with ex-post monetary compensation among TSOs based on actual costs of procuring the reserves. While options (1) and (2) can be integrated into the modeling frameworks directly, option (3) might be harder to implement under current market structures and European regulations.

Energinet's current dimensioning methodology forecasts the FRR considering netting and sharing to some degree, but lacks full coordination with the other TSOs. The exact methodology will be explored in the following section.

4 Current Methodology for Dynamic FRR Dimensioning

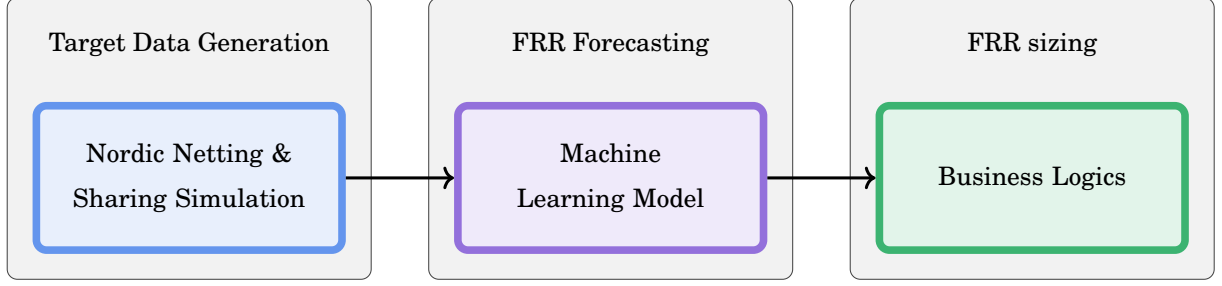


Figure 9: Current Energinet Methodology for Dynamic Dimensioning of mFRR

To harvest the benefits of a *dynamic* procurement approach Energinet has implemented a methodology consisting of three independent steps, seen in Figure 9.

First, an optimization model is executed to simulate the amount of FRR that would perfectly cover system imbalances and a potential reference incident in the Nordic region, accounting for netting and sharing possibilities among Nordic TSOs. The resulting simulations serve as target variable for the data set used in the FRR forecasting step.

The machine learning model in the FRR forecasting step is developed within a probabilistic framework. Consequently, Energinet obtains an hourly FRR probability distribution for each hour of the upcoming day. From these distributions, a quantile is selected to reflect the desired risk level. These quantiles, combined with a set of business logics, determines the final aFRR and mFRR capacity to procure in the capacity market for each hour of the next 24-hour period. The resulting time series is then submitted to the Nordic Market Management System (Nordic MMS) and used in the market auctions. Each of these steps is described in detail in the following subsections.

4.1 Nordic Netting and Sharing simulation

The Nordic Netting and Sharing (NNAS) methodology is described in Figure 10.

The first model, the *Control Area Netting*, uses historical normal imbalances (NI) and Available Transfer Capacity (ATC) as inputs. Its objective is to minimize imbalances within each Control Area (CA) by netting imbalances between Load Frequency Control (LFC) areas (e.g., DK1 and DK2 for Denmark, or SE1–SE4 for Sweden) whenever possible, and by redistributing any remaining imbalances across LFC areas. The decision variables of the problem are therefore the flow between LFC areas, and the imbalances at each LFC area. Constraint (5b) ensures that the initial imbalance of a node is always equal to the current imbalance considering in- and outflows to the LFC area. The set of constraints (5d) manages the ATC of interconnectors. The resulting decision variables from the *CA Netting Model* are the leftover imbalance at each LFC area and the residual ATC values, which serve as input to the *Nordic Netting Model*.

The second model follows a similar structure and logic in the constraints, but allows netting of imbalances across control areas, e.g. Norway is allowed to net its imbalances with Sweden and so forth. However, the imbalance in each LFC area is constrained by an *Imbalance Target*, which is defined by the share of

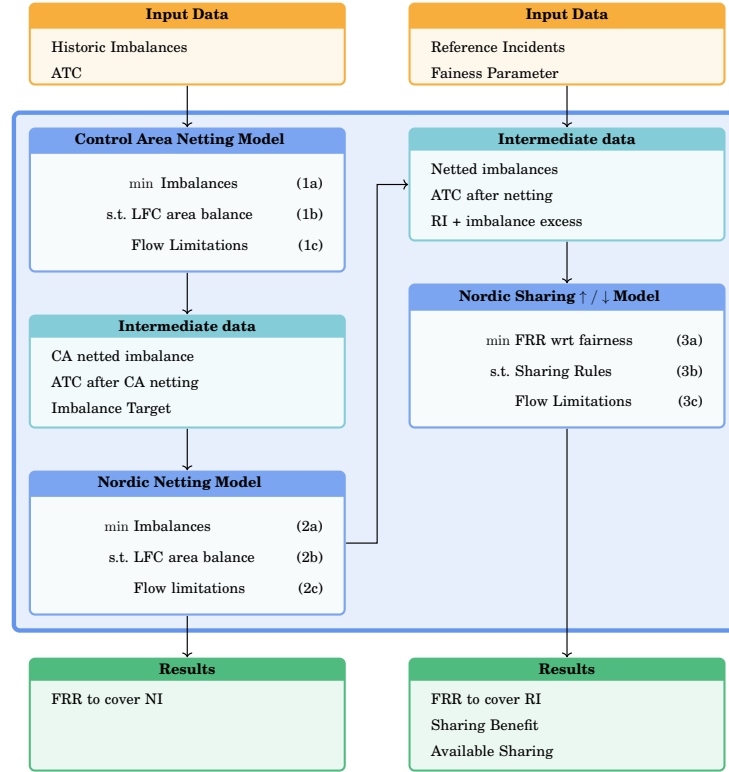


Figure 10: Nordic Netting and Sharing Simulation workflow. The boxes indicated as models are all linear programs.

‘pollution’ each LFC area brings with respect to its initial imbalance. E.g. if DK1 had a positive imbalance, and the total imbalance in the Nordic region after control area netting was positive, the *Imbalance Target* will be the share of the initial imbalances compared to the total imbalance. If the sum of imbalances was instead negative, then DK1’s share would be set to zero, as it was *helping* the overall system.

The final step in the NNAS simulation is the Nordic Sharing Model $\uparrow/\downarrow$, which consists of two separate models optimizing up- and down-regulation needs separately (as indicated by the arrows in the title in Figure 10). These models use outputs from the previous step and incorporate two additional inputs: the Reference Incident (RI) and a *Fairness Parameter*. The RI, defined per LFC area, determines the target for the FRR need. If excess imbalance from the previous step would support system stability (e.g., positive RI combined with negative imbalance excess), the FRR target can be reduced accordingly. The Fairness Parameter acts as a hyperparameter controlling how strongly the objective function penalizes uneven sharing among TSOs. This is crucial because well-connected areas can bear a disproportionate sharing burden, reflected in higher reserve procurement for the TSO, but ultimately reducing overall system reserve needs. Energinet is actively using different fairness measures, with another method being setting an upper bound on the sharing per interconnector. This allows all TSOs to have the potential of benefitting the same, while having a worst-case guarantee. The average reduction in reserve needs as a function of sharing limits is shown in Figure 11.

The full mathematical formulation of each model is provided in Appendix A.



Figure 11: The graphs show the average FRR need when allowing different levels of sharing per interconnector.

4.2 FRR Forecasting

Energinet employs Bayesian Additive Regression Trees (BART) [29] as the core forecasting model for dynamic dimensioning. BART is an ensemble of regression trees that produces a full predictive distribution conditional on the input features, rather than a point estimate, which allows credible intervals and arbitrary quantiles to be extracted directly from the forecast. A mathematical formulation of BART is provided in Appendix B; the merits and limitations of this modelling choice are discussed in section 5.

In the current Energinet approach, two distinct BART models are implemented: one for predicting the reserves needed to cover the reference incident and normal imbalances and one for predicting voluntary bids. The first model is trained against the FRR target value generated by the NNAS simulation, whereas the voluntary-bids model uses historical voluntary bids as its target. The two resulting probability distributions are convolved into a single distribution representing the final FRR probability using a Monte Carlo sampling approach. The forecast pipeline is illustrated in Figure 12.

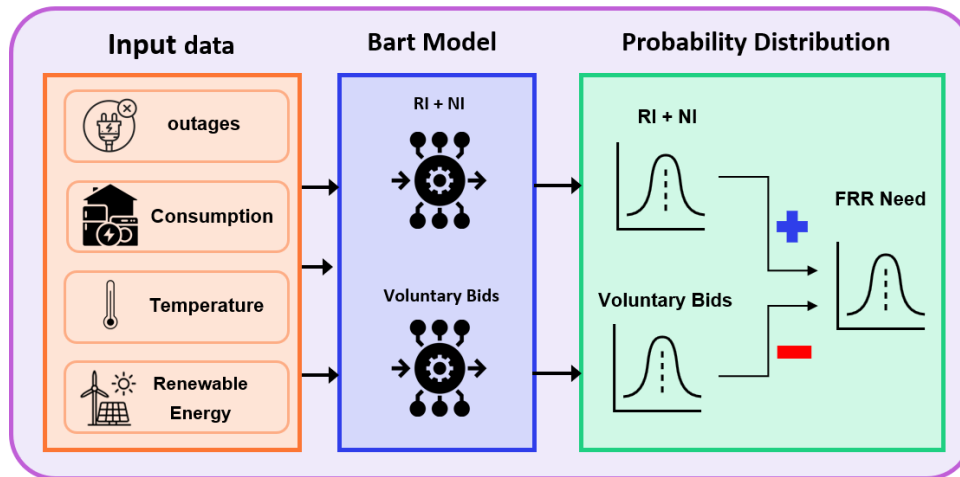


Figure 12: Energinet's current FRR forecasting setup: two BART models, one for imbalances (trained on the NNAS simulation output), one for voluntary bids, are convolved into a single FRR distribution.

4.3 Sizing of FRR Procurement

Energinet has currently implemented dynamic dimensioning only for the mFRR product, while a fixed amount of aFRR is maintained. Specifically, Energinet procures 100 MW upward and 100 MW downward for DK1, and between 40 MW and 90 MW in both directions for DK2, for all hours.

Once the forecasted FRR probability distribution is obtained, Energinet selects a quantile that reflects its predefined risk tolerance. The mFRR capacity to be procured for each hour is then computed as the chosen quantile of the FRR distribution minus the fixed aFRR procurement,

$$\text{mFRR} = q_{\alpha}(\mu_{\text{FRR}}) - \text{aFRR}, \quad \alpha \in (0, 1), \quad (4)$$

where μ_{FRR} is the probability distribution computed by the BART pipeline, α represents the risk level, and $q_{\alpha}(\cdot)$ is the corresponding quantile of μ_{FRR} . To mitigate the risk of model overfitting and extreme values, Energinet applies fixed lower and upper bounds to the calculated mFRR requirement, limiting the procurement range to between 250 MW and 350 MW.

5 Beyond the current method

Energinet has employed dynamic dimensioning of mFRR for an extended period, and while the approach has proven successful from both business and engineering perspectives, several challenges and open questions remain. In particular, model transparency is crucial for broader adoption among Nordic TSOs: BART predictions are produced by aggregating numerous regression trees, which makes the resulting probabilistic forecasts difficult to explain in terms of individual input features or operational scenarios. This lack of interpretability can hinder trust and acceptance, especially when compared to more transparent approaches such as linear or rule-based models.

To address these concerns, Energinet might explore a shift in methodology, by reversing the order of steps in the current framework (see Figure 9), so that the forecasting element becomes the initial step. The revised approach aims to train forecasting models directly on historical data rather than simulated target values. These models would generate probabilistic forecasts for key parameters such as Available Transfer Capacity (ATC), Reference Incident (RI), imbalance, and voluntary bids. The forecasts would then serve as inputs to an optimization framework that determines the total FRR requirement. This optimization would be formulated as a multi-area problem, accounting for netting and sharing opportunities, and would intrinsically incorporate the desired risk level for each TSO. Modeling this step with a transparent optimization model (e.g., linear programming) ensures interpretability and trust in the results. The optimization model would then yield a FRR need for each TSO. In the last step, each TSO would then be able to split the total FRR volume into aFRR and mFRR components according to their own preferences, allowing for both flexibility and coordinated dimensioning. The updated methodology is depicted in Figure 13.



Figure 13: Potential methodology update for the common nordic dynamic dimensioning of FRR

5.1 Forecasting system parameters

In the proposed methodology the forecast becomes *component-based*: the components feeding into the Joint Netting and Sharing Mechanism (JNSM) are forecasted individually, in a common output format, while each TSO is free to choose its own forecasting technique. Energinet could, with relatively low effort, adapt its existing BART pipeline to produce these component forecasts. The validation of these forecasted parameters remains an open question. The main advantage over the current approach is more reliable forecasting, since historical data for each component is readily available, rather than depending on a simulated training target.

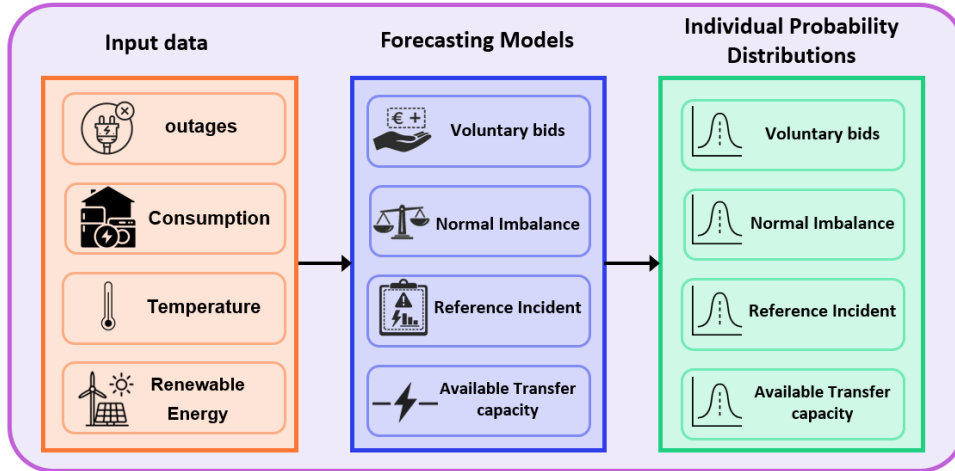


Figure 14: Proposed methodology for forecasting input parameters to the common FRR optimization framework

5.2 Joint Netting & Sharing Mechanism

The current NNAS framework at Energinet is used for simulating the data needed to train FRR forecasting models. While the underlying linear optimization problems are transparent and interpretable, this transparency is largely lost when the outputs are used to train black-box machine learning models.

The proposed JNSM aims to retain a linear optimization structure while explicitly accounting for uncertainty in key input parameters, thereby allowing risk preferences to be represented directly within the optimization problem and, if desired, differentiated across TSOs. Figure 15 shows the proposed frame-

work, where the forecasted probability distributions of different system parameters act as uncertain inputs to an optimization model. Several formulations could capture this uncertainty, including stochastic, chance-constrained, and risk-based approaches.

Multi-area reserve coordination has been studied previously. [30] examine cross-border allocation and sharing without explicit robustness to uncertainty. [31] propose a chance-constrained two-stage formulation that allows reserve requirements to be satisfied only in a predefined share of scenarios, but restricted to a single control area within one TSO. [32] extend this to all Nordic control areas, focusing on how the availability of (simulated) information affects outcomes. In contrast, the JNSM is intended as an *operational* framework that integrates data-driven forecasts of uncertain inputs and delivers concrete, actionable decisions for joint netting and sharing across multiple TSOs. The exploration of alternative formulations constitutes natural future work at Energinet.

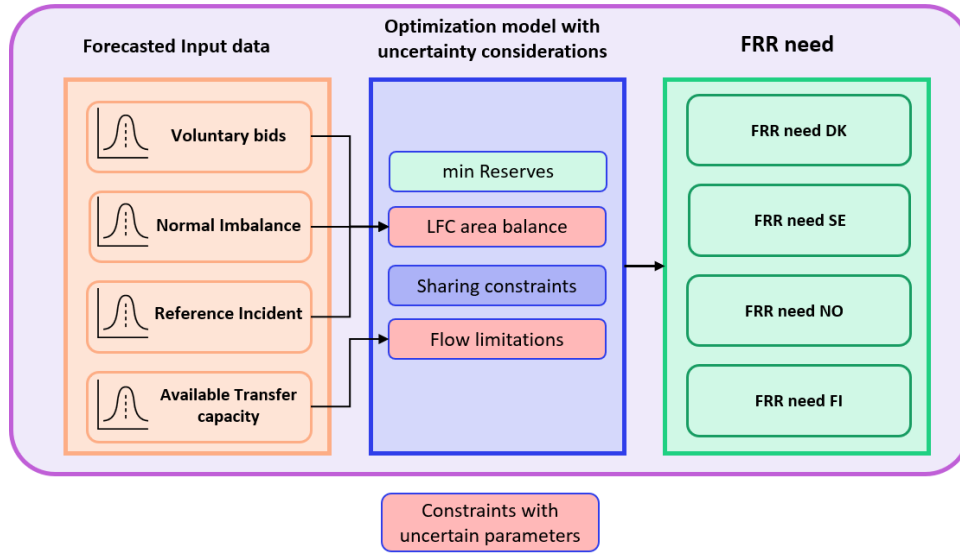


Figure 15: Proposed methodology for Joint Netting and Sharing model for the coordinated dimensioning of FRR in the Nordics. It builds on an estimate of uncertain input parameters and deals with decision making under uncertainty by applying a robust optimization modelling framework.

5.3 Rule-based FRR splitting

In the current Energinet methodology, Energinet targets the total FRR need, rather than the individual components of aFRR and mFRR. In a more coordinated framework, as envisioned with the JNSM, it is theorized that keeping the target as FRR will be a beneficial approach as it allows each TSO to have their own custom method for splitting FRR into aFRR and mFRR. This is relevant, since an *optimal* approach can highly depend on the system characteristics, bilateral agreements between the TSO and ancillary services providers etc.

6 Conclusion and future work

Energinet's initial dynamic dimensioning approach has been live for one year and is regarded as a success from an operational perspective. Several research directions remain for further developing the methodology:

1. **Quantifying *fair* reserve sharing.** A starting point is to survey existing notions of fair resource sharing in the literature and apply them to a Nordic case study.
2. **Joint multi-TSO optimisation under uncertainty.** This entails identifying the key uncertain parameters, estimating their uncertainty, and building and empirically validating an optimisation framework that coordinates FRR dimensioning across TSOs while being robust to that uncertainty.
3. **Splitting FRR into aFRR and mFRR.** How the total FRR need should be allocated between the two products in a coordinated framework remains an open question.

7 Abbreviations

ATC Available Transfer Capacity

BSP Balancing Service Provider

FFR Fast Frequency Reserve

FRR Frequency Restoration Reserve

FCR Frequency Containment Reserve

FCR-D Frequency Containment Reserve - Disturbance

FCR-N Frequency Containment Reserve - Normal

FRR Frequency Restoration Reserves

aFRR Automatic Frequency Restoration Reserves

mFRR Manual Frequency Restoration Reserves

MARI Manually Activated Reserves Initiative

MTU Market Time Unit

VB Voluntary Bids

RI Reference Incident

NI Normal Imbalances

EAM Energy Activation Market

CM Capacity Market

LFC Load Frequency Control

TSO Transmission System Operator

PICASSO Platform for the International Coordination of Automated Frequency Restoration and Stable System Operation

JNSM Joint Netting and Sharing Mechanism

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A Nordic Netting and Sharing model definition

The NNAS simulation introduced in section 5 consists of three sequential linear programs: *Control Area Netting*, *Nordic Netting* and *Nordic Sharing*. The three models share a common notation, summarised below:

- $\mathcal{Z}$ Set of all LFC zones z .
- $\mathcal{C}$ Set of interconnectors, containing ordered pairs (u, v) .
- $\mathcal{N}_z$ Set of interconnectors incident to zone z . E.g.,
 $\mathcal{N}_{DK1} = \{(DK1, DK2), (DK1, NO2), (DK1, SE3)\},$
 $\mathcal{N}_{DK2} = \{(DK2, DK1), (DK2, SE4)\}.$
- $\mathcal{A}$ Set of control areas (e.g., Denmark, Sweden, Norway, Finland).
- $\mathcal{P}$ Set of all (unordered) zone pairs (u, v) in the Nordic synchronous area.
- $\mathcal{P}_a^{CA}$ Set of (unordered) zone pairs (u, v) within control area $a \in \mathcal{A}$.
- $\mathcal{N}_z^{CA}$ Subset of $\mathcal{N}_z$ restricted to interconnectors inside the control area of z .

In all three models, interconnector flows are signed scalars and antisymmetry is enforced explicitly so that a flow from u to v equals the negative of the flow from v to u . The absolute-value terms appearing in the netting objectives act on the residual *imbalances* (which can take either sign) and serve to drive them towards zero. Throughout this appendix, a superscript star (e.g., i_z^* , $f_{u,v}^*$) denotes the value taken by a decision variable at an optimal solution of the corresponding optimisation problem; starred quantities from one stage are passed as inputs to the next.

A.1 Control Area Netting

The first stage nets imbalances within each control area independently. Inputs are the per-zone normal imbalances I_z and the per-interconnector available transfer capacities $ATC_{u,v}$. The model is:

$$\min_{\Theta^{CA}} \sum_{z \in \mathcal{Z}} |i_z| + \sum_{a \in \mathcal{A}} \sum_{(u,v) \in \mathcal{P}_a^{CA}} |i_u - i_v| \quad (5a)$$

subject to

$$i_z = I_z + \sum_{v: (z,v) \in \mathcal{N}_z^{CA}} f_{z,v} \quad \forall z \in \mathcal{Z} \quad (5b)$$

$$f_{u,v} = -f_{v,u} \quad \forall (u, v) \in \mathcal{C} \quad (5c)$$

$$-ATC_{v,u} \leq f_{u,v} \leq ATC_{u,v} \quad \forall (u, v) \in \mathcal{C} \quad (5d)$$

$$i_z \in \mathbb{R} \quad \forall z \in \mathcal{Z} \quad (5e)$$

The model determines the residual imbalance i_z in each zone and the netting flows $f_{u,v}$ on each interconnector, with decision-variable set $\Theta^{CA} = \{i_z, f_{u,v}\}$. The first term in the objective (5a) minimises the absolute residual imbalance in every zone, whereas the second term penalises imbalance differences

between LFC zones *belonging to the same control area*, thereby promoting an even distribution of any unavoidable residual imbalance within a control area. Constraint (5b) is the per-zone imbalance balance: the residual imbalance equals the initial imbalance plus the net inflow over the intra-CA interconnectors $\mathcal{N}_z^{CA}$. Cross-CA interconnectors are excluded at this stage, so netting cannot occur between, e.g., DK1 and SE3. Since interconnectors are bi-directional, flows are allowed to take either sign and antisymmetry is enforced through (5c); capacity limits in either direction follow from (5d). The outputs of this stage are the post-CA-netting imbalances $I_z^{\text{After CA netting}} := i_z^*$ and the residual capacities $ATC_{u,v}^{CA} := ATC_{u,v} - f_{u,v}^*$, which are passed to the Nordic Netting model.

A.2 Nordic Netting

The second stage nets imbalances *across* control areas, using the residual imbalances and ATCs from the previous stage as inputs. The model is:

$$\min_{\Theta^{NN}} \sum_{z \in \mathcal{Z}} |i_z^{\text{netted}}| + \sum_{z \in \mathcal{Z}} |i_z^{\text{netted}} - I_z^{\text{Target}}| \quad (6a)$$

subject to

$$i_z^{\text{netted}} = I_z^{\text{After CA netting}} + \sum_{v: (z,v) \in \mathcal{N}_z} f_{z,v}^{\text{netted}} \quad \forall z \in \mathcal{Z} \quad (6b)$$

$$f_{u,v}^{\text{netted}} = -f_{v,u}^{\text{netted}} \quad \forall (u,v) \in \mathcal{C} \quad (6c)$$

$$-ATC_{v,u}^{CA} \leq f_{u,v}^{\text{netted}} \leq ATC_{u,v}^{CA} \quad \forall (u,v) \in \mathcal{C} \quad (6d)$$

$$i_z^{\text{netted}} \in \mathbb{R} \quad \forall z \in \mathcal{Z} \quad (6e)$$

where the per-zone *imbalance target* encodes a fairness rule:

$$I_z^{\text{Target}} = \begin{cases} I^{\text{Nordic}} \cdot \frac{I_z^{\text{after CA netting}}}{I^{\text{Nordic sum in dominating direction}}} & \text{if } \text{sign}(I_z^{\text{before netting}}) = \text{sign}(I^{\text{Nordic}}), \\ 0 & \text{if } \text{sign}(I_z^{\text{before netting}}) \neq \text{sign}(I^{\text{Nordic}}), \end{cases} \quad \forall z \in \mathcal{Z}. \quad (7)$$

The decision variables are the netted imbalances i_z^{netted} and the cross-zonal netting flows $f_{u,v}^{\text{netted}}$, i.e., $\Theta^{NN} = \{i_z^{\text{netted}}, f_{u,v}^{\text{netted}}\}$. The first term in the objective (6a) drives the absolute residual imbalance in every zone towards zero, exploiting cross-CA interconnectors that were not available in the previous stage. The second term anchors the per-zone outcome to the imbalance target I_z^{Target} defined in (7): a zone whose imbalance was sign-aligned with the aggregate Nordic imbalance (i.e., it was *contributing to* the system imbalance) is assigned a residual proportional to its share of the dominating direction, while a zone whose imbalance was already *helping* the overall system is assigned a target of zero. This prevents helping zones from being pushed into a residual imbalance for the sole purpose of improving another zone's metric. Constraint (6b) is the imbalance balance over the full neighbour set $\mathcal{N}_z$, while (6c)–(6d) again enforce flow antisymmetry and respect the residual ATC headroom $ATC_{u,v}^{CA}$ inherited from the CA Netting stage. The outputs of this stage are the post-netting imbalances $i_z^{\text{netted}*}$, used to derive the imbalance excess $R_z^{\text{imbalance excess}}$ that enters the Sharing model, and the further reduced residual capacities $ATC_{u,v}^{NN} := ATC_{u,v}^{CA} - f_{u,v}^{\text{netted}*}$.

A.3 Nordic Sharing model

$$\min_{\Theta} \sum_z r_z^{\text{RI}} + K \sum_{(u,v) \in \mathcal{P}} |s_u^{\text{benefit}} - s_v^{\text{benefit}}| - 0.01 \sum_z s_z^{\text{available}} \quad (8a)$$

subject to

$$r_z^{\text{RI}} + s_z^{\text{available}} \geq R_z^{\text{RI isolated}} \quad \forall z \in \mathcal{Z} \quad (8b)$$

$$s_z^{\text{benefit}} = R_z^{\text{RI isolated}} - r_z^{\text{RI}} \quad \forall z \in \mathcal{Z} \quad (8c)$$

$$s_z^{\text{available}} = \sum_{v:(v,z) \in \mathcal{N}_z} f_{z,v,z} \quad \forall z \in \mathcal{Z} \quad (8d)$$

$$\sum_{v:(z,v) \in \mathcal{N}_z} f_{x,z,v} \leq r_z^{\text{RI}} \quad \forall z \in \mathcal{Z}, \forall x \in \mathcal{Z}, x \neq z \quad (8e)$$

$$f_{z,u,v} = -f_{z,v,u} \quad \forall (u,v) \in \mathcal{C}, \forall z \in \mathcal{Z} \quad (8f)$$

$$-ATC_{v,u}^{\text{NN}} \leq f_{z,u,v} \leq ATC_{u,v}^{\text{NN}} \quad \forall (u,v) \in \mathcal{C}, \forall z \in \mathcal{Z} \quad (8g)$$

$$R_z^{\text{RI min}} \leq r_z^{\text{RI}} \leq \max\{R_z^{\text{RI min}}, R_z^{\text{RI iso}}\} \quad \forall z \in \mathcal{Z} \quad (8h)$$

$$0 \leq s_z^{\text{benefit}} \leq R_z^{\text{RI iso}} \quad \forall z \in \mathcal{Z} \quad (8i)$$

$$0 \leq s_z^{\text{available}} \leq \sum_{v:(z,v) \in \mathcal{N}_z} ATC_{v,z} \quad \forall z \in \mathcal{Z} \quad (8j)$$

The decision variables are the locally procured RI reserves r_z^{RI} , the cross-zonal sharing flows $f_{z,u,v}$ (denoting the contribution over interconnector (u,v) towards covering a potential reference incident in zone z), and the auxiliary quantities $s_z^{\text{available}}$ and s_z^{benefit} , i.e., $\Theta = \{r_z^{\text{RI}}, f_{z,u,v}, s_z^{\text{available}}, s_z^{\text{benefit}}\}$. The objective (8a) minimises the total procured RI reserves and penalises unequal sharing benefits between zones through the term $K \sum_{(u,v) \in \mathcal{P}} |s_u^{\text{benefit}} - s_v^{\text{benefit}}|$, where K acts as a fairness hyperparameter. The last term, with a small negative weight on $\sum_z s_z^{\text{available}}$, is a tie-breaker that favours solutions making fuller use of the available sharing potential. Constraint (8b) requires that locally procured reserves together with reserves available from other zones cover the isolated reference-incident requirement $R_z^{\text{RI isolated}}$, while (8c) and (8d) define the auxiliary variables: s_z^{benefit} captures the reduction in local procurement enabled by sharing, and $s_z^{\text{available}}$ aggregates the inflow of sharing reserves into zone z . Constraint (8e) enforces that a zone cannot export more reserves to support a peer zone x than the amount of RI reserves it has procured locally, and (8g) bounds sharing flows by the residual capacities $ATC_{u,v}^{\text{NN}}$ inherited from the Nordic Netting stage. Bounds (8h)–(8j) restrict the procurement and sharing variables to physically meaningful ranges. The isolated reference-incident requirement is computed as $R_z^{\text{RI isolated}} = \max(0, \text{RI}_z - R_z^{\text{imbalance excess}})$, where RI_z is the reference incident in zone z and $R_z^{\text{imbalance excess}}$ is the imbalance excess inherited from the Nordic Netting stage. The formulation is generic and applies to both up- and down-regulation, with the sign of the relevant flow direction reversed between the two cases.

B Mathematical formulation of BART

Let $\{(x_i, y_i)\}_{i=1}^n$ denote the observed data, where $x_i \in \mathbb{R}^p$ are predictors and $y_i \in \mathbb{R}$ the response. Bayesian Additive Regression Trees (BART) [29] model the data as

$$y_i = f(x_i) + \epsilon_i, \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad (9)$$

where $f(\cdot)$ is represented as a sum of m regression trees:

$$f(x) = \sum_{j=1}^m g(x; T_j, M_j). \quad (10)$$

Here T_j denotes the structure of the j -th binary tree (its internal splitting rules) and $M_j = \{\mu_{j,1}, \dots, \mu_{j,b_j}\}$ collects the parameters at its b_j terminal (leaf) nodes. The function $g(x; T_j, M_j)$ returns the leaf parameter $\mu_{j,\ell}$ associated with the unique leaf ℓ that x falls into under T_j , so each individual tree is a piecewise-constant function of x .

Bayesian regularisation is imposed through three independent priors on (T_j, M_j, σ^2) . The tree-structure prior is specified implicitly by the probability that a node at depth d is non-terminal,

$$P_{\text{split}}(d) = \alpha (1 + d)^{-\beta}, \quad \alpha \in (0, 1), \beta \geq 0, \quad (11)$$

together with uniform priors on the splitting variable and split point at each internal node; the parameters (α, β) encode a strong preference for shallow trees. Conditional on T_j , the leaf parameters are taken to be i.i.d.

$$\mu_{j,\ell} \mid T_j \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\mu^2), \quad \ell = 1, \dots, b_j, \quad (12)$$

where σ_μ is calibrated relative to the (centred and scaled) range of y , typically as $\sigma_\mu = 0.5/(k\sqrt{m})$ for some $k > 0$, so that each individual tree contributes only a small share of the overall fit. The error variance follows a conjugate inverse-gamma prior,

$$\sigma^2 \sim \text{Inv-Gamma}\left(\frac{\nu}{2}, \frac{\nu\lambda}{2}\right), \quad (13)$$

with hyperparameters (ν, λ) chosen so that the prior places most of its mass below a data-driven estimate of the residual variance. Together, these priors prevent any single tree from dominating the ensemble, so that the sum of trees behaves as a flexible but well-regularised regression function.

Posterior inference proceeds via a Bayesian backfitting Markov Chain Monte Carlo (MCMC) sampler [29]: at each iteration, (T_j, M_j) is updated for $j = 1, \dots, m$ in turn, conditional on the partial residuals

$$R_{i,j} = y_i - \sum_{k \neq j} g(x_i; T_k, M_k), \quad (14)$$

followed by a Gibbs update for σ^2 . The sampler targets the joint posterior

$$p(T_1, M_1, \dots, T_m, M_m, \sigma^2 \mid y, X), \quad (15)$$

which induces a posterior distribution over the regression function $f(x)$. For a new input x^* , point predictions are obtained as the posterior mean $\hat{f}(x^*) = \mathbb{E}[f(x^*) \mid y, X]$, and predictive uncertainty is summarised by credible intervals or arbitrary quantiles of the posterior predictive distribution of $y^* \mid x^*, y, X$, the latter being the property exploited for risk-aware reserve dimensioning.

